

CSE321 Assignment 02

Research and **Explore** why modern operating systems use multiple page sizes instead of simply replacing conventional pages with huge pages, and investigate how huge/superpages can improve address translation performance while introducing new memory management challenges. Analyze the following key areas:

- **TLB Reach and Multiple Page Sizes**
- **Huge/Superpage Formation and Management**

Report Requirements:

Write a comprehensive analysis focusing on the following key factors:

Address Translation and Performance:

- **Explain** how page size affects TLB reach and investigate why applications with large memory footprints may experience significant TLB misses when conventional pages are used.
- **Analyze** how huge/superpages increase TLB coverage and how this can reduce logical to physical address translation overhead.
- **Investigate** why supporting multiple page sizes may provide better performance than simply increasing the default page size for the entire system.
- **Analyze** the relationship among page size, number of page table entries, TLB misses, page table walks, and memory access performance.

Huge/Superpage Management and Trade-offs:

- **Investigate** how an operating system can create a huge/superpage from multiple conventional pages through page promotion/coalescing, and explain why physical memory contiguity becomes important during this process.
- **Analyze** how memory fragmentation can prevent the operating system from forming huge/superpages even when sufficient total free memory exists.
- **Analyze** situations in which using huge/superpages may provide little benefit or may even negatively affect system performance.

Study Resources:

- Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Operating Systems: Three Easy Pieces (OSTEP), 1.10 Edition, Chapters: 18, 19, 20
- Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Operating System Concepts, 10th Edition: Chapters 9, 10
- [Talluri, M., Kong, S. I., Hill, M. D., and Patterson, D. A., "Tradeoffs in Supporting Two Page Sizes," ISCA, 1992.](#)
- [Romer, T. H., Ohlrich, W. H., Karlin, A. R. and Bershad, B. N., "Reducing TLB and Memory Overhead Using Online Superpage Promotion," ISCA, 1995.](#)
- [Navarro, J., Iyer, S., Druschel, P. and Cox, A., "Practical, Transparent Operating System Support for Superpages," OSDI, 2002.](#)
- [Kwon, Y., Yu, H., Peter, S., Rossbach, C. J. and Witchel, E., "Ingens: Huge Page Support for the OS and Hypervisor."](#)

Submission guidelines: Can be found in the submission form given below.

Submission link:

<https://forms.gle/cxAc1taC1C9WRiT6>

Deadline: 8 September 2026, 11:00 PM

(Submissions outside the Google Form will be automatically rejected.)